

LECTURE 8: ENGINEERING ETHICS

Core Principle: Legal compliance = floor, not ceiling. Prioritize public safety.

Case Studies:

- **Case: Citicorp Center (LeMessurier)**
 - → Problem: Hidden weak bolts + 40% more wind force = collapse risk.
 - → Lesson: Disclosing mistake & fixing (secret welding) enhanced reputation; didn't destroy it.
- **Case: GE & GM (Headlights/Lamps)**
 - → Problem: Proprietary safety tech vs public safety.
 - → Lesson: Duty to share safety advances industry-wide, even if costing competitive edge.
- **Case: NHTSA vs. IIHS**
 - → Problem: NHTSA (binding/frontal) vs IIHS (nonprofit/offset). Car passes legal but fails offset.
 - → Lesson: Don't ignore better safety data just because design meets legal minimum.

3 Excuses (Impediments to Ethics):

1. "Not my problem" → False. Unsafe products raise prices/taxes/insurance for all.
 2. "If I don't, someone else will" → Being first to build harm isn't an achievement.
 3. "I can't foresee everything" → Build foresight into design; if mistake happens, step up & fix.
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LECTURE 9: AI BIAS & FAIRNESS

Bias Definition: Prejudice for/against thing/person/group vs another.

Causes in Tech: 1. Incomplete Data 2. Human Perceptions (tags).

Bias Across Software Lifecycle:

- Requirements → Whose needs are defined?
- Data Collection → Selection/time/observer biases.
- Modeling → Bias-variance trade-off.
- Testing → Subgroup evaluation (not just overall avg).
- Deployment → Automated & complacency bias.

Types of Bias:

- **Sociological:** Racism, Sexism, Ageism (excluding elderly ui), Religious Prejudice.
- **Implicit:** Unconscious stereotypes. Often contradict conscious beliefs.
 - *Affinity Bias* → Favoring people similar to us.
- **Cognitive (Mental Shortcuts):** (Hook: "FAAO")
 - **False Consensus** → Overestimate how normal our own views are.
 - **Anchoring** → Over-reliance on 1st piece of info.
 - **Attribution Error** → Judge others by character, not circumstances.
 - **Optimism Bias** → Believe negative risks apply less to you than others.
- **Systemic/Human-Machine Biases:**
 - **Confirmation Bias:** 1. Hypothesize → 2. Seek favorable info (ignore rest) → 3. Publish.
 - **Complacency Bias:** Trust machine > human.
 - **Automated Bias:** Trust machine implicitly → Stop monitoring → "Omission errors" (the miss) go undetected.

Data & Statistical Biases:

- **Data Collection Biases:**

- Selection → Not randomized/representative.
- Time Interval → Too narrow time frame.
- Observer → Humans collecting data judge via cognitive bias.
- Misclassification → Mislabeled data.

- **Statistical Biases:**

- Outliers → Drastically different points.
- Underfitting → Removed too many variables; model fails to learn.
- Overestimating Rule → Initially include as many variables as possible.
- Uncorrelated Data → Must remove (e.g., ice cream sales & drowning).

Modeling Bias (The Trade-off):

- **Bias** = Matches training set exactly.
- **Variance** = Flexibility with different data.
- **Underfit** → High Bias + Low Variance (Too simple).
- **Overfit** → Low Bias + High Variance (Memorized data, fails new data).
- **Goal** → Low Bias + Low Variance (Minimizes "Generalization Error").

Fairness Metrics & Ethics:

- **Demographic Parity** → Equal outcome rates across groups (regardless of qual).
- **Equalized Odds** → Equal error rates (false +/-) across groups.
- **The Impossibility Result** → Mathematically impossible to satisfy both if base rates differ. Choosing is an ethical decision, must explicitly document.

Engineer's 6 Responsibilities:

1. Audit Data
2. Test Subgroups
3. Diversify Team

4. Document/Disclose limitations
5. Don't Automate Blindly (Human in loop)
6. Speak Up.

Case Studies:

- **Case: Gender Shades (Facial Recog)**
 - → Problem: Trained on white males. Failed on women/POC.
 - → Causes: Selection Bias + Modeling Bias + Sociological Bias.
 - **Case: COMPAS (Recidivism Algorithm)**
 - → Problem: False high-risk for Black defendants; false low-risk for White defendants.
 - → Lesson: Equal overall accuracy != Equalized Odds. Kept as "trade secret" (unauditable).
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LECTURE 10: RECOMMENDER SYSTEMS & MODERATION

Recommender Systems: Maximize engagement → Spreads lies 6.5x faster than truth. **Moderation Methods:**

1. Take Down
2. Demote (pushes to bottom)
3. Flag (marks false + context).

Case Studies:

- **Case: "Innocence of Muslims" (YouTube)**
 - → Problem: False video → Violent protests/deaths. YT refused takedown.
 - → Generalizability: Removing inciting violence = generalizable. Removing *all* lies = not generalizable (destroys utility).

- → Utilitarian: YT consequences matter, even if rioters caused immediate harm.
 - → Free Speech: 1st Amendment restricts Govt, not YT.
 - → Autonomy Principle: Bystanders hurt in riot didn't consent (Violation).
 - → Compromise: Flag + reliable sources (satisfies utility & speech).
 - **Case: Teen Mental Health vs Social Media**
 - → Problem: Spikes in depression linking to SM use.
 - → Autonomy: Banning kids != violation (no right to access). Kids cannot give *informed consent*.
 - → Lesson: Utilitarian benefits (fun/edu) cannot outweigh autonomy violation (harm) for kids.
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LECTURE 11: LLMs & GANS

LLM Mechanics:

- Word associations.
- Uses *Spatial Word Embedding* (words = points in high-dim space, reduced via singular value decomposition).
- No actual understanding;

GANs (Generative Adversarial Networks):

- **Generator** (makes fakes) vs **Discriminator** (tries to catch fakes) → hyper-realistic output.
- **ChatGPT Flaws:** Pre-trained (no live learning), Hallucinations (invents fake facts/images).

Ethics of AI Writing:

- **Generalization Argument 1 (Implied Promise):** Author promises original thought. GPT breaks this.
- **Generalization Argument 2 (Garbage Loop):** AI trains on human data. If humans stop writing → AI eats AI data → "Garbage in,

garbage out".

- **Exceptions:** Boilerplate/routine text. Assisted writing (if edited/fact-checked).
- **Utilitarian Argument:** Writing builds literacy/reasoning. Using GPT deprives society of skilled thinkers.

Misuse & Jailbreaking:

- **Jailbreaking:** Tricking AI to bypass safety rules.
 - **Techniques:** Role-play, multi-step reasoning, obfuscation/encoding, prompt injection.
 - **Lesson:** Security researchers do this responsibly to patch flaws. Engineers must design robust defenses.
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LECTURE 12: IP & GENERATIVE AI

The AI IP Problem:

- **Scraping:** Downloading bulk public data without permission/pay.
- **The Garbage Loop Test:** Scrape free → nobody pays creators → creation stops → nothing left to scrape. (Root issue = funding problem, e.g., NYT Lawsuit).
- **IP Nature:** Ideas cannot be owned. Copying != Taking (Non-exclusive). Patent = trade (disclose for 20yr monopoly).

TRIPS Agreement: WTO treaty (1994). Sets global *minimum* baseline protections (Floor, not ceiling).

The 4 Kinds of Intellectual Property: 1. Copyright

- → Protects: *Expression* of idea (code, text, art). NOT the idea/algorithm itself.
- → Acquired: Automatically on creation.
- → Lifespan: Life + 50 yrs (TRIPS min). US: Life + 70, or 95

(Corporate - e.g., Steamboat Willie).

- → Fair Use: Allowed for criticism, news, teaching, research.
- → Software: Yes (source code).

2. Trademark

- → Protects: Brand identity (logo/name identifying source).
- → Acquired: Commercial use or registration.
- → Lifespan: Indefinite (if in commercial use).
- → Rule: Class-specific (Apple records vs Apple computers).

3. Patent

- → Protects: New method, device, material.
- → Acquired: Apply + Exam + Public Disclosure.
- → Lifespan: 20 years.
- → Exclusions (Cannot Patent): Pure ideas (math), Nature (DNA), Biz methods, "Look and feel".
- → Software: Yes, ONLY if it improves computer functionality (Not just "doing X on a computer").

4. Trade Secret

- → Protects: Secret info giving commercial advantage (e.g., Coca-Cola formula).
- → Acquired: Keeping it secret.
- → Lifespan: Indefinite (until leaked/discovered).
- → Rule: Legal to reverse engineer or figure out independently.
Illegal to steal docs. NDAs used for employees